

2 Fig. 2.1 is a simplified diagram of the human circulatory system.

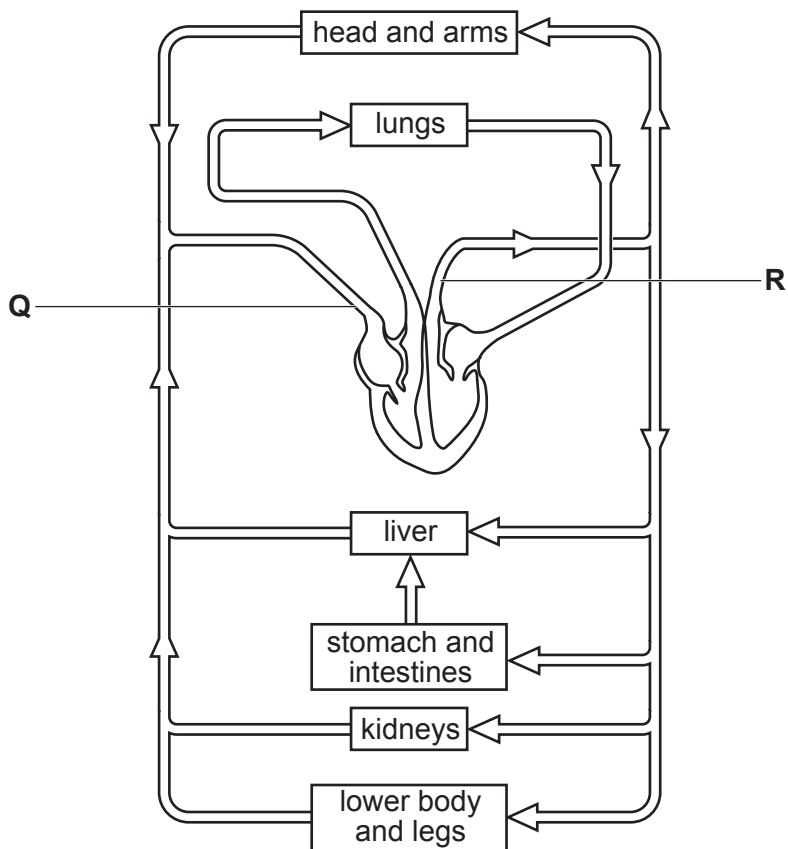


Fig. 2.1

(a) (i) State why the human circulatory system is described as a closed, double circulation.

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..... [2]

(ii) Identify blood vessel **Q** and blood vessel **R**, shown in Fig. 2.1.

Q

R

[1]

- (iii) Blood leaving the heart passes through valves before it enters blood vessel **R**.

Describe the differences between the structure of blood vessel **Q** and the structure of blood vessel **R**. Do **not** refer to valves in your answer.

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..... [3]

- (b) Blood plasma plays an important role in the transport of molecules such as antibodies.

Scientists discovered that some of the antibodies in the blood plasma of sharks have a different structure to the antibodies found in human blood plasma.

Fig. 2.2 shows the structure of an antibody molecule found in the blood plasma of a shark.

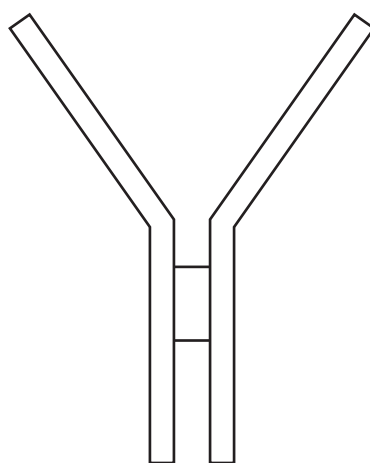


Fig. 2.2

- (i) State how the quaternary structure of a human antibody molecule differs from the quaternary structure of the shark antibody molecule shown in Fig. 2.2.

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- (ii) Human antibodies are used in the treatment of some forms of cancer. However, the antibodies injected into the bloodstream can only reach a small percentage of the cancer cells that form the cancerous tumour.

Shark antibodies are smaller than human antibodies. Scientists are researching the possibility of injecting shark antibodies into the bloodstream to treat cancerous tumours in humans.

Suggest how using the smaller shark antibodies may be more effective in reaching a greater percentage of cancer cells than human antibodies and lead to greater success at treating cancer.

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..... [2]

- (iii) Antibodies can also be used in the prevention of infectious diseases.

Explain how injection of antibodies into the bloodstream can protect a person from disease after infection by a pathogen.

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..... [3]

[Total: 12]